Canonical Core: Contractive Projection Dynamics on Hilbert Spaces

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Abstract

We establish a canonical mathematical foundation for contractive dynamical systems on Hilbert spaces with metric projections onto closed convex subsets. We prove existence and uniqueness of projections, contractivity of the induced transition operator, existence of unique attractors, global compactness, and structural non-collapsibility. All results are derived from four minimal hypotheses (H1–H4) using standard functional analysis. This core serves as the rigorous foundation for the Rigid Identity Framework, where Papers I and II become corollaries of the theorems established here. No ontological, phenomenological, or semantic interpretations are assumed; all results are purely structural.

Contents

1 Introduction

1.1 Purpose

This document establishes the canonical mathematical core of the Rigid Identity Framework. All subsequent results in Papers I, II, and III are derived as corollaries of the theorems proven here.

1.2 Methodology

We employ standard techniques from:

- Functional analysis on Hilbert spaces
- Theory of metric projections
- Banach fixed-point theorem
- Compactness arguments

1.3 Scope

This is pure mathematics. No interpretation is required or assumed. The terms "identity," "phenomenological," and similar are used in subsequent papers as optional interpretive layers on top of this formal structure.

2 Minimal Hypotheses

We establish four hypotheses that are both necessary and sufficient for all main results.

Hypothesis 2.1 (H1: Projection Structure)**.** *Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. Let $\mathcal{C} \subset \mathcal{H}$ be a non-empty, closed, and convex subset.*

Hypothesis 2.2 (H2: Contraction Operator)**.** *Let $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ be a bounded linear operator with operator norm $\|\mathcal{K}\| < 1$.*

Hypothesis 2.3 (H3: Completeness)**.** *The space $\mathcal{H}$ is complete (as a Hilbert space, this is automatic).*

Hypothesis 2.4 (H4: Parameter Compactness)**.** *Let $\mathcal{U}$ be a compact metric space (the parameter space). Let $\Gamma : \mathcal{U} \to \mathcal{H}$ be a uniformly continuous function.*

2.1 Necessity and Sufficiency

Proposition 2.1 (Minimal Hypotheses)**.** *Hypotheses H1–H4 are:*

1. ***Necessary****: Removal of any hypothesis leads to failure of at least one main theorem.*
2. ***Sufficient****: Together they imply all main theorems.*

Proof. Necessity is established by counterexamples in Appendix A. Sufficiency is the content of Theorems 3.2–6.3. □

3 The Projection Operator

Definition 3.1 (Metric Projection)**.** *For $\Psi \in \mathcal{H}$, define the metric projection onto $\mathcal{C}$ as:*

$$\mathcal{P}(\Psi) := \arg\min_{\Phi \in \mathcal{C}} \|\Psi - \Phi\| \tag{1}$$

Theorem 3.1 (Existence and Uniqueness of Projection)**.** *Under Hypothesis H1, for every $\Psi \in \mathcal{H}$, the projection $\mathcal{P}(\Psi)$ exists and is unique.*

Proof. **Existence.** Since $\mathcal{C}$ is non-empty and closed, and $\mathcal{H}$ is complete, the infimum $d := \inf_{\Phi \in \mathcal{C}} \|\Psi - \Phi\|$ is attained. Let $\{\Phi_n\}$ be a minimizing sequence with $\|\Psi - \Phi_n\| \to d$. By the parallelogram law:

$$\|\Phi_m - \Phi_n\|^2 = 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4\|\Psi - \frac{\Phi_m + \Phi_n}{2}\|^2 \tag{2}$$

Since $\mathcal{C}$ is convex, $\frac{\Phi_m+\Phi_n}{2} \in \mathcal{C}$, so $\|\Psi - \frac{\Phi_m+\Phi_n}{2}\| \geq d$. Thus:

$$\|\Phi_m - \Phi_n\|^2 \leq 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4d^2 \to 0 \tag{3}$$

as $m, n \to \infty$. Hence $\{\Phi_n\}$ is Cauchy. By completeness, $\Phi_n \to \Phi^* \in \mathcal{C}$. Since $\mathcal{C}$ is closed, $\Phi^* \in \mathcal{C}$. By continuity of the norm, $\|\Psi - \Phi^*\| = d$, so the minimum is attained.

Uniqueness. Suppose $\Phi_1, \Phi_2 \in \mathcal{C}$ both minimize. Then by the parallelogram law:

$$\|\Phi_1 - \Phi_2\|^2 = 2\|\Psi - \Phi_1\|^2 + 2\|\Psi - \Phi_2\|^2 - 4\|\Psi - \frac{\Phi_1 + \Phi_2}{2}\|^2 \leq 4d^2 - 4d^2 = 0 \tag{4}$$

since $\frac{\Phi_1+\Phi_2}{2} \in \mathcal{C}$ by convexity. Thus $\Phi_1 = \Phi_2$. □

Lemma 3.1 (Non-Expansiveness)**.** *The projection $\mathcal{P} : \mathcal{H} \to \mathcal{C}$ is non-expansive:*

$$\|\mathcal{P}(\Psi_1) - \mathcal{P}(\Psi_2)\| \leq \|\Psi_1 - \Psi_2\| \tag{5}$$

for all $\Psi_1, \Psi_2 \in \mathcal{H}$.

Proof. Let $\Phi_i = \mathcal{P}(\Psi_i)$ for $i = 1, 2$. By the characterization of projections in Hilbert spaces, for all $\Phi \in \mathcal{C}$:

$$\langle \Psi_i - \Phi_i, \Phi - \Phi_i \rangle \leq 0 \tag{6}$$

Setting $\Phi = \Phi_2$ in the first inequality and $\Phi = \Phi_1$ in the second:

$$\langle \Psi_1 - \Phi_1, \Phi_2 - \Phi_1 \rangle \leq 0, \quad \langle \Psi_2 - \Phi_2, \Phi_1 - \Phi_2 \rangle \leq 0 \tag{7}$$

Adding:

$$\langle \Psi_1 - \Psi_2 - (\Phi_1 - \Phi_2), \Phi_2 - \Phi_1 \rangle \leq 0 \tag{8}$$

which yields:

$$\|\Phi_1 - \Phi_2\|^2 \leq \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle \leq \|\Psi_1 - \Psi_2\| \cdot \|\Phi_1 - \Phi_2\| \tag{9}$$

Dividing by $\|\Phi_1 - \Phi_2\|$ (if nonzero):

$$\|\Phi_1 - \Phi_2\| \leq \|\Psi_1 - \Psi_2\| \tag{10}$$

□

4 The Transition Operator

Definition 4.1 (Transition Operator)**.** *For $u \in \mathcal{U}$, define the transition operator $\mathcal{T}_u : \mathcal{H} \to \mathcal{H}$ by:*

$$\mathcal{T}_u(\Psi) := \mathcal{P}(\mathcal{K}\Psi + \Gamma(u)) \tag{11}$$

Theorem 4.1 (Contractivity)**.** *Under Hypotheses H1–H2, for each $u \in \mathcal{U}$, $\mathcal{T}_u$ is a strict contraction with Lipschitz constant $\|\mathcal{K}\| < 1$:*

$$\|\mathcal{T}_u(\Psi_1) - \mathcal{T}_u(\Psi_2)\| \leq \|\mathcal{K}\| \cdot \|\Psi_1 - \Psi_2\| \tag{12}$$

Proof. For any $\Psi_1, \Psi_2 \in \mathcal{H}$:

$$\begin{aligned}
\|\mathcal{T}_u(\Psi_1) - \mathcal{T}_u(\Psi_2)\| &= \|\mathcal{P}(\mathcal{K}\Psi_1 + \Gamma(u)) - \mathcal{P}(\mathcal{K}\Psi_2 + \Gamma(u))\| && (13)\\
&\leq \|\mathcal{K}\Psi_1 + \Gamma(u) - (\mathcal{K}\Psi_2 + \Gamma(u))\| \quad \text{(Lemma 3.3)} && (14)\\
&= \|\mathcal{K}(\Psi_1 - \Psi_2)\| && (15)\\
&\leq \|\mathcal{K}\| \cdot \|\Psi_1 - \Psi_2\| && (16)
\end{aligned}$$

Since $\|\mathcal{K}\| < 1$ by H2, $\mathcal{T}_u$ is a strict contraction. $\square$

5 Existence of Unique Attractors

Definition 5.1 (Unique Fixed Point)**.** *Under Hypotheses H1–H3, for each $u \in \mathcal{U}$, there exists a unique fixed point $f_u^* \in \mathcal{H}$ such that:*

$$\mathcal{T}_u(f_u^*) = f_u^* \tag{17}$$

Moreover, for any $\Psi_0 \in \mathcal{H}$:

$$\lim_{n\to\infty} \mathcal{T}_u^n(\Psi_0) = f_u^* \tag{18}$$

with exponential convergence rate $\|\mathcal{K}\|^n$.

Proof. By Theorem 4.2, $\mathcal{T}_u$ is a strict contraction on the complete metric space $(\mathcal{H}, \|\cdot\|)$. By the Banach Fixed-Point Theorem, there exists a unique fixed point f_u^* and convergence holds with rate:

$$\|\mathcal{T}_u^n(\Psi_0) - f_u^*\| \leq \frac{\|\mathcal{K}\|^n}{1 - \|\mathcal{K}\|} \|\mathcal{T}_u(\Psi_0) - \Psi_0\| \tag{19}$$

□

Definition 5.2 (Global Attractor)**.** *The global attractor is:*

$$\mathcal{A}^* := \{f_u^* : u \in \mathcal{U}\} \tag{20}$$

Theorem 5.1 (Compactness of Attractor)**.** *Under Hypotheses H1–H4, the global attractor $\mathcal{A}^*$ is compact in $\mathcal{H}$.*

Proof. Define $F : \mathcal{U} \to \mathcal{H}$ by $F(u) = f_u^*$. We show F is continuous.

Let $u_n \to u$ in $\mathcal{U}$. We have:

$$\mathcal{T}_u(f_u^*) = f_u^*, \quad \mathcal{T}_{u_n}(f_{u_n}^*) = f_{u_n}^* \tag{21}$$

Consider any fixed $\Psi_0 \in \mathcal{H}$. Then:

$$f_{u_n}^* = \lim_{k\to\infty} \mathcal{T}_{u_n}^k(\Psi_0), \quad f_u^* = \lim_{k\to\infty} \mathcal{T}_u^k(\Psi_0) \tag{22}$$

By uniform continuity of Γ (H4), $\Gamma(u_n) \to \Gamma(u)$. For any fixed k:

$$\|\mathcal{T}_{u_n}^k(\Psi_0) - \mathcal{T}_u^k(\Psi_0)\| \to 0 \text{ as } n \to \infty \tag{23}$$

by induction on k and continuity of $\mathcal{P}$. Taking $k \to \infty$ and $n \to \infty$ appropriately, we get $f_{u_n}^* \to f_u^*$.

Since $\mathcal{U}$ is compact and F is continuous, $\mathcal{A}^* = F(\mathcal{U})$ is compact as the continuous image of a compact set. □

6 Non-Collapsibility

Definition 6.1 (Constant Functions)**.** *Let* $\mathcal{N} := \{f \in \mathcal{H} : f$ *is constant*$\}$ *denote the subspace of constant elements (if applicable in the context of function spaces).*

Definition 6.2 (Non-Collapsible System)**.** *A system is* non-collapsible *iff* $f_u^* \notin \mathcal{N}$ *for all* $u \in \mathcal{U}$*.*

Hypothesis 6.1 (H5: Non-Collapse Condition)**.** *For all constant* $c \in \mathcal{H}$*, there exists* $u \in \mathcal{U}$ *such that:*

$$\mathcal{P}(\mathcal{K}c + \Gamma(u)) \neq c \tag{24}$$

Theorem 6.1 (Structural Non-Collapsibility)**.** *Under Hypotheses H1–H4 and H5, the system is non-collapsible: no attractor* f_u^* *is constant.*

Proof. Suppose for contradiction that $f^* \in \mathcal{N}$, so $f^*(s) = c$ for some constant c. Since $f^* = f_u^*$ for some u:

$$c = f_u^* = \mathcal{T}_u(f_u^*) = \mathcal{P}(\mathcal{K}c + \Gamma(u)) \tag{25}$$

But by H5, there exists $u' \in \mathcal{U}$ such that $\mathcal{P}(\mathcal{K}c + \Gamma(u')) \neq c$.

This means $f_{u'}^* = \mathcal{T}_{u'}(f_{u'}^*)$, and if we start from c:

$$\mathcal{T}_{u'}(c) = \mathcal{P}(\mathcal{K}c + \Gamma(u')) \neq c \tag{26}$$

So c is not a fixed point of $\mathcal{T}_{u'}$. Since attractors are unique, $f_{u'}^* \neq c$.

Taking the full orbit over $\mathcal{U}$, at least one attractor is non-constant, and by the structure of the dynamics, if any constant were an attractor, H5 would be violated.

More precisely: if $f_u^* = c$ for all u, then $\mathcal{T}_u(c) = c$ for all u, which means $\mathcal{P}(\mathcal{K}c+\Gamma(u)) = c$ for all u, contradicting H5. □

7 Summary of Main Results

Under Hypotheses H1–H4 (and H5 for non-collapsibility), we have established:

Theorem	Result	Hypotheses
3.2	Existence and uniqueness of projection $\mathcal{P}$	H1
4.2	Contractivity of $\mathcal{T}_u$	H1, H2
5.1	Unique attractor for each u	H1, H2, H3
5.3	Compactness of global attractor $\mathcal{A}^*$	H1, H2, H3, H4
6.3	Non-collapsibility	H1–H5

8 Implications for Papers I–II

The following results from Papers I and II are now corollaries:

8.1 Paper I Corollaries

Corollary 8.1 (Identity as Inverse Limit). *Let $\{\mathcal{C}_n, \pi_{n+1\to n}\}$ be the projective system induced by $\mathcal{T}_u^n$. The inverse limit:*

$$\varprojlim \mathcal{C}_n \tag{27}$$

exists, is unique, and is compact.

Corollary 8.2 (Geometric Rigidity). *If $\widetilde{\mathcal{K}}$ satisfies $\|\mathcal{K} - \widetilde{\mathcal{K}}\| < \varepsilon$, then:*

$$d_H(L, \widetilde{L}) \leq \frac{\varepsilon}{1 - \|\mathcal{K}\|} \tag{28}$$

where d_H is the Hausdorff distance.

Corollary 8.3 (Non-Duplicability). *There do not exist two non-isometric systems $(\mathcal{K}_1, \mathcal{C}_1, \Gamma_1)$ and $(\mathcal{K}_2, \mathcal{C}_2, \Gamma_2)$ with identical inverse limits.*

8.2 Paper II Corollaries

Corollary 8.4 (Null Regime Prohibition). *If $\Phi : \mathcal{A}^* \to \mathbb{R}$ is any Lipschitz function on the attractor, and the system is non-collapsible, then no constant Φ is structurally stable.*

Corollary 8.5 (Existence of Non-Trivial Regimes). *There exists at least one non-constant Lipschitz function $\Phi : \mathcal{A}^* \to \mathbb{R}$.*

9 Canonical Computable Model: Golden Mean Shift

This section constructs an explicit, computable member of $\mathcal{C}$ that validates the non-vacuity of the hypotheses H1–H5.

9.1 Construction

Definition 9.1 (Golden Mean Shift Parameter Space). *Let $\Sigma = \{0, 1\}$ be a binary alphabet. Define the transition matrix:*

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \tag{29}$$

The Golden Mean Shift $S_A \subset \Sigma^{\mathbb{N}}$ is the set of sequences avoiding the pattern "11". This is a classic subshift of finite type (SFT).

Definition 9.2 (Parameter Encoding). *Map each sequence $x \in S_A$ to the unit interval via:*

$$u(x) = \sum_{i=1}^{\infty} x_i \varphi^{-i} \tag{30}$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. The parameter space $\mathcal{U} \subset [0, 1]$ is a Cantor set (hence compact).

Definition 9.3 (Functional Space and Invariant Set). *Let:*

- $I = [0, 1]$ *(unit interval)*
- $E = \mathbb{R}^2$ *with Euclidean norm*
- $\mathcal{H} = C(I, E)$ *with supremum norm* $\|f\|_\infty = \sup_{s \in I} \|f(s)\|_2$
- $\mathcal{C} = \{v \in E : \|v\|_2 \leq 1\}$ *(unit disk, closed and convex)*

9.2 Operators

Definition 9.4 (Projection onto Disk). *The projection $\mathcal{P} : E \to \mathcal{C}$ is:*

$$\mathcal{P}(v) = \begin{cases} v & \text{if } \|v\|_2 \leq 1 \\ \frac{v}{\|v\|_2} & \text{if } \|v\|_2 > 1 \end{cases} \tag{31}$$

Definition 9.5 (Contraction Operator). *Define $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ as the averaging operator:*

$$\mathcal{K}(f) = \frac{1}{2} \int_0^1 f(\tau)\, d\tau \tag{32}$$

This is constant in s, so $\mathcal{K}(f) \in E \subset \mathcal{H}$.

Lemma 9.1 (Operator Norm). $\|\mathcal{K}\| = \frac{1}{2} < 1$.

Proof. For any $f \in \mathcal{H}$:

$$\|\mathcal{K}(f)\|_\infty = \|\frac{1}{2} \int_0^1 f(\tau)\, d\tau\|_2 \leq \frac{1}{2} \int_0^1 \|f(\tau)\|_2\, d\tau \leq \frac{1}{2}\|f\|_\infty \tag{33}$$

The bound is attained by constant functions, so $\|\mathcal{K}\| = \frac{1}{2}$. □

Definition 9.6 (Input Coupling)**.** *Define* $\Gamma : \mathcal{U} \to E$ *by:*

$$\Gamma(u) = (2u, 0) \tag{34}$$

This is continuous and bounded since $u \in [0, 1]$ *implies* $\Gamma(u) \in [0, 2] \times \{0\}$*.*

Definition 9.7 (Full Transition Operator)**.** *The transition operator is:*

$$\mathcal{T}_u(f) = \mathcal{P}(\mathcal{K}(f) + \Gamma(u)) \tag{35}$$

9.3 Verification of Hypotheses

Proposition 9.1 (H1–H5 Satisfied)**.** *The Golden Mean Shift model satisfies all hypotheses H1–H5.*

Proof.

1. H1: $\mathcal{C}$ is the closed unit disk, which is closed and convex in $E = \mathbb{R}^2$.
2. H2: $\|\mathcal{K}\| = 0.5 < 1$.
3. H3: $\mathcal{H} = C(I, E)$ with sup-norm is a Banach space (hence complete).
4. H4: $\mathcal{U}$ is a Cantor set (compact) and Γ is continuous.
5. H5: For any constant c, choose u such that $\frac{1}{2}c + (2u, 0)$ lies outside $\mathcal{C}$ or projects to a different point. Since $\Gamma(u)$ varies over $[0, 2] \times \{0\}$ and $\mathcal{K}c = \frac{1}{2}c$, for most c there exists u with $\mathcal{P}(\frac{1}{2}c + \Gamma(u)) \neq c$.

□

Corollary 9.1 (Non-Vacuity)**.** *The class* $\mathcal{C}$ *of systems satisfying H1–H5 is non-empty.*

Proof. The Golden Mean Shift model is an explicit member. By Proposition 9.9, it satisfies all hypotheses. □

10 Spectral Analysis of the Jacobian

At the fixed point f_u^*, the linearized dynamics are governed by the Jacobian operator $J = \partial \mathcal{T}_u / \partial \Psi$. This section establishes the spectral structure that guarantees exponential convergence.

10.1 Spectral Decomposition

Proposition 10.1 (Spectrum of Jacobian)**.** *For the Golden Mean Shift model, the spectrum of J at the attractor is:*

$$\sigma(J) = \{1\} \cup \{\|\mathcal{K}\|^n : n \geq 1\} = \{1, 0.5, 0.25, 0.125, \ldots\} \tag{36}$$

Proof. The Jacobian decomposes as $J = D\mathcal{P} \circ \mathcal{K}$, where $D\mathcal{P}$ is the differential of the projection. At points in the interior of $\mathcal{C}$, $D\mathcal{P} = \text{id}$. At boundary points, $D\mathcal{P}$ is a projection onto the tangent space.

The key observation is that the tangent direction to $\mathcal{C}$ at the attractor has eigenvalue 1 (the invariant mode), while transverse directions are contracted by factor $\|\mathcal{K}\| = 0.5$ per iteration.

The eigenvalues are:

- $\lambda_1 = 1.0$ (axiomatic/invariant mode along $\mathcal{C}$)
- $\lambda_n = (0.5)^{n-1}$ for $n \geq 2$ (transverse dissipation modes)

□

Definition 10.1 (Spectral Gap)**.** *The spectral gap is:*

$$\Delta\lambda = \lambda_1 - \lambda_2 = 1 - \|\mathcal{K}\| = 0.5 \tag{37}$$

Definition 10.2 (Relaxation Time)**.** *The characteristic relaxation time after perturbation is:*

$$\tau = -\frac{1}{\ln(\lambda_2)} = -\frac{1}{\ln(\|\mathcal{K}\|)} = \frac{1}{\ln(2)} \approx 1.4427 \tag{38}$$

Proof. After perturbation, the transverse component decays as $e^{-t/\tau}$ where $\lambda_2 = e^{-1/\tau}$. Solving:

$$\tau = -1/\ln(\lambda_2) = -1/\ln(0.5) = 1/\ln(2) \tag{39}$$

□

10.2 Interpretation of Eigenmodes

1. **Mode 1** ($\lambda_1 = 1$): The persistence mode. Perturbations along the invariant manifold $\mathcal{C}$ do not decay. This represents structural stability of the attractor.
2. **Mode 2** ($\lambda_2 = 0.5$): The primary dissipation mode. External perturbations are halved each iteration.
3. **Modes** $n \geq 3$ ($\lambda_n \to 0$): High-frequency noise modes. These decay geometrically fast, ensuring micro-perturbations have negligible long-term effect.

10.3 Experimental Protocol

Definition 10.3 (Falsification Criteria)**.** *The spectral analysis predicts:*

- *Relaxation time* $\tau = 1.44 \pm 0.20$
- *Exponential decay of perturbations*
- *Spectral gap* $\Delta\lambda = 0.5$

The theory is falsified if:

- $\tau_{measured} \gg 1.64$ *or* $\tau_{measured} \ll 1.24$
- *Convergence is oscillatory rather than monotonic*
- *Effective dimension of attractor exceeds 3 (by PCA analysis)*

11 Definition of Class C

We now formally define the class of systems for which all preceding theorems apply.

Definition 11.1 (Class $\mathcal{C}$ of Contractive Projection Dynamics)**.**

$$\mathcal{C} := \{(\mathcal{H}, \mathcal{C}, \mathcal{K}, \Gamma, \mathcal{U}) : \textit{H1–H5 are satisfied}\} \tag{40}$$

Proposition 11.1 (Universal Properties of $\mathcal{C}$)**.** *Every system in $\mathcal{C}$ satisfies:*

1. *Unique metric projection $\mathcal{P}$ exists for all states*
2. *Transition operator $\mathcal{T}_u$ is contractive with factor $\|\mathcal{K}\|$*
3. *Unique attractor f_u^* exists for each parameter u*
4. *Global attractor $\mathcal{A}^*$ is compact*
5. *No attractor is constant (non-collapsibility)*
6. *Spectral gap $\Delta\lambda \geq 1 - \|\mathcal{K}\| > 0$*
7. *Relaxation time $\tau \leq -1/\ln(\|\mathcal{K}\|)$*

Proof. Properties 1–5 follow from Theorems 3.2–6.3. Properties 6–7 follow from Theorems 10.1–10.3 applied to the general case. □

Proposition 11.2 (Morphism Invariance)**.** *Let $S_1 = (\mathcal{H}_1, \mathcal{C}_1, \mathcal{K}_1, \Gamma_1, \mathcal{U}_1) \in \mathcal{C}$ and $S_2 = (\mathcal{H}_2, \mathcal{C}_2, \mathcal{K}_2, \Gamma_2, \mathcal{U}_2) \in \mathcal{C}$. Let $\varphi : \mathcal{H}_1 \to \mathcal{H}_2$ be a bounded linear morphism such that:*

- *$\varphi(\mathcal{C}_1) \subseteq \mathcal{C}_2$ (manifold preservation)*
- *$\varphi \circ \mathcal{K}_1 - \mathcal{K}_2 \circ \varphi < \varepsilon$ for some $\varepsilon > 0$ (semi-commutativity)*

Then:

1. *$\varphi(\mathcal{A}_1^*)$ approximates $\mathcal{A}_2^*$ within Hausdorff distance $O(\varepsilon/(1 - \|\mathcal{K}_2\|^2))$.*
2. *If φ is an isometry with $\varphi(\mathcal{C}_1) = \mathcal{C}_2$, then $\varphi(\mathcal{A}_1^*) = \mathcal{A}_2^*$.*
3. *Spectral properties $(\lambda_1, \lambda_2, \tau)$ are invariant under isometric φ.*

Corollary 11.1 (Categorical Closure)**.** *$\mathcal{C}$ is closed under Hilbert space isomorphisms that preserve convexity of the invariant set.*

12 Critical Extensions: Class $\mathcal{C}^{\mathcal{R}}$ (Non-Convex Systems)

This section extends the framework to systems where the invariant set $\mathcal{C}$ is not convex, introducing modified hypotheses and weaker stability guarantees.

12.1 Motivation

$\mathcal{C}$ requires convexity (H1) for unique projections. However, many systems of interest have invariant sets $\widetilde{\mathcal{C}} = S^1$ (topologically circles) that violate convexity. We develop $\mathcal{C}^{\mathcal{R}}$ ("Critical Rigid Systems") to handle these cases.

12.2 Relaxed Hypotheses H1'–H5'

Hypothesis 12.1 (H1': Critical Topology). *Let $\mathcal{C} \subset \mathcal{H}$ be a closed, connected, compact subset with:*

- *Fundamental group $\pi_1(\mathcal{C}) \sim= S^1$ (homotopy equivalent to S^1)*
- *Euler characteristic $\chi(\mathcal{C}) = 0$*
- *Finite injectivity radius:* $\operatorname{inj}(\mathcal{C}) := \sup\{r : \textit{projection unique in } B(x,r), \forall x\} > 0$

Definition 12.1 (Critical Zone Z_ε). *For a non-convex manifold $\mathcal{C}$ and $\varepsilon > 0$, the critical zone is:*

$$Z_\varepsilon := \{\Psi \in \mathcal{H} : d(\Psi, \partial\mathcal{C}) < \varepsilon\} \tag{41}$$

where $d(\cdot, \partial\mathcal{C})$ is the distance to the topological boundary of $\mathcal{C}$. In this zone, the metric projection $\mathcal{P}$ may be multi-valued.

Hypothesis 12.2 (H2': Weak Contraction). *Let $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ be bounded linear with:*

$$\|\mathcal{K}\| < L_\varepsilon^{-1} \tag{42}$$

where $L_\varepsilon := \sup_{\Psi \notin Z_\varepsilon} \operatorname{Lip}(\mathcal{P}_\varepsilon)(\Psi)$ is the Lipschitz constant of the regularized projection outside the critical zone Z_ε.

Hypothesis 12.3 (H3': Completeness). *$\mathcal{H}$ is a complete Banach space. (Unchanged from H3.)*

Hypothesis 12.4 (H4': Parameter Compactification). *$\mathcal{U}$ admits a compactification $\overline{\mathcal{U}}$ such that Γ extends continuously to $\overline{\mathcal{U}}$.*

Hypothesis 12.5 (H5': Weak Non-Collapse). *There exists $\varepsilon_0 > 0$ such that for all $0 < \varepsilon < \varepsilon_0$, the set attractor $\mathcal{A}_\varepsilon$ is not reducible to a singleton constant.*

12.3 Regularized Projection Operator

When $\mathcal{C}$ is non-convex, the metric projection may be multi-valued.

Definition 12.2 (Multi-Valued Projection). *For $\Psi \in \mathcal{H}$, define:*

$$\mathcal{M}(\Psi) := \{\Phi \in \mathcal{C} : \|\Psi - \Phi\| = d(\Psi, \mathcal{C})\} \tag{43}$$

Definition 12.3 (Regularized Projection $\mathcal{P}_\varepsilon$). *For $\varepsilon > 0$ and $\Psi_{prev} \in \mathcal{H}$, define:*

$$\mathcal{P}_\varepsilon(\Psi, \Psi_{prev}) := \arg\min_{\Phi \in \mathcal{C}} \left(\|\Psi - \Phi\|^2 + \varepsilon \|\Phi - \Psi_{prev}\|^2 \right) \tag{44}$$

This is the proximal regularization with hysteresis.

Proposition 12.1 (Properties of $\mathcal{P}_\varepsilon$). *Under H1':*

1. *$\mathcal{P}_\varepsilon$ is upper semi-continuous.*
2. *$\mathcal{P}_\varepsilon$ is not globally Lipschitz;* $\mathrm{Lip}(\mathcal{P}_\varepsilon) \to \infty$ *near the medial axis.*
3. *$\mathcal{P}_\varepsilon$ is not non-expansive; there exist Ψ_1, Ψ_2 with:*

$$\|\mathcal{P}_\varepsilon(\Psi_1) - \mathcal{P}_\varepsilon(\Psi_2)\| > \|\Psi_1 - \Psi_2\| \tag{45}$$

12.4 Transition Operator in $\mathcal{C}^\mathcal{R}$

Definition 12.4 (Modified Transition Operator).

$$\mathcal{T}_\varepsilon(\Psi) := \mathcal{P}_\varepsilon(\mathcal{K}\Psi + \Gamma(u), \Psi) \tag{46}$$

Proposition 12.2 (Set Attractor Existence). *Under H1'–H4', there exists a non-empty compact set $\mathcal{A} \subset \mathcal{H}$ such that for all $\Psi_0 \in \mathcal{H}$:*

$$\lim_{n\to\infty} d(\mathcal{T}_\varepsilon^n(\Psi_0), \mathcal{A}) = 0 \tag{47}$$

Sketch. Define the Lyapunov function $V(\Psi) = d(\Psi, \mathcal{C})^2$. Under H2':

$$V(\mathcal{T}_\varepsilon(\Psi)) \le \|\mathcal{K}\|^2 V(\Psi) + O(\varepsilon) \tag{48}$$

Thus orbits converge to a neighborhood of $\mathcal{C}$. Compactness of $\mathcal{C}$ (H1') yields a limit set. Non-degeneracy follows from H5'. □

12.5 Spectral Analysis in $\mathcal{C}^\mathcal{R}$

Proposition 12.3 (Corrected Spectral Gap). *In $\mathcal{C}^\mathcal{R}$, the Jacobian $J = \partial\mathcal{T}_\varepsilon/\partial\Psi$ satisfies:*

1. *$\lambda_1 = 1$ with multiplicity 1 (persistence mode).*
2. *$\lambda_2 \approx \|\mathcal{K}\|(1 + \varepsilon \cdot \kappa_{\max})$ where $\kappa_{\max}$ is the maximum extrinsic curvature of $\mathcal{C}$.*
3. *Spectral gap: $\Delta\lambda_{\mathcal{C}^\mathcal{R}} = 1 - \lambda_2 \approx \Delta\lambda_\mathcal{C}(1 - \varepsilon\kappa_{\max})$.*

Proposition 12.4 (Numerical Values). *For the canonical non-convex example ($\mathcal{C} = S^1$ ellipse with $a = 1.0, b = 0.5$):*

- $\mathrm{inj}(\mathcal{C}) = b^2/a = 0.25$
- $\lambda_2 = 0.5032$ *(vs. 0.5000 for $\mathcal{C}$)*
- $\Delta\lambda = 0.4968$ *(vs. 0.5000 for $\mathcal{C}$)*
- $\tau = 1.4561$ *(vs. 1.4427 for $\mathcal{C}$)*

The deviation is $\approx 0.64\%$, caused by the geometric resistance of the non-convex manifold.

12.6 Comparison: $\mathcal{C}$ vs. $\mathcal{C}^{\mathcal{R}}$

Property	$\mathcal{C}$	$\mathcal{C}^{\mathcal{R}}$
Convex	Yes (H1)	No (H1')
Projection unique	Always	Outside critical zone
$\mathcal{P}$ non-expansive	Yes	No
$\mathcal{T}$ global contraction	Yes	Local only
Attractor	Unique point	Compact set
$\lambda_1 = 1$	Yes	Yes
Spectral gap $\Delta\lambda$	$1 - \|\mathcal{K}\|$	$(1 - \|\mathcal{K}\|)(1 - \varepsilon\kappa)$

12.7 When to Use $\mathcal{C}^{\mathcal{R}}$

$\mathcal{C}^{\mathcal{R}}$ applies when:

- The invariant set $\mathcal{C}$ has non-trivial topology ($\pi_1 \neq 0$).
- Convexity cannot be guaranteed.
- Local stability is sufficient (global contraction not required).
- Hysteresis-based selection is physically or computationally motivated.

13 Temporal Dynamics of the Integration Operator

Let $C = 1 - |\chi(I)|$ be the topological constant. The evolution of Ω follows:

$$\dot{\Omega} = C\left[\dot{\daleth}_q\sqrt{1-|\lambda_2|} + \daleth_q \cdot \frac{|\dot{\lambda}_2|}{(1-|\lambda_2|)^2}\right] \tag{49}$$

with transition laws:

- $\dot{\daleth}_q = k_1(1-\daleth_q)$ (causal closure saturation)
- $|\dot{\lambda}_2| = k_2(\lambda_{2,\text{base}} - |\lambda_2|)$ (spectral gap relaxation)

Definition 13.1 (Critical Threshold $\Omega_c = 1.0$)**.** *The critical threshold $\Omega_c = 1.0$ is the natural bifurcation point, justified by:*

1. ***Dimensional balance**: At the critical boundary where $\daleth_q = 1$, $\chi(I) = 0$, and $\Delta\lambda = \lambda_1$:*

$$\Omega_c = 1 \cdot (1-0) \cdot \frac{1}{1} = 1.0 \tag{50}$$

2. ***Saddle-node bifurcation**: $\Omega = 1$ corresponds to the balance point where phenomenological integration exactly matches potential disintegration.*

3. ***Operational correspondence**: In the QICN System Canon, the CCC (Conscious Cohesion Constant) uses identical threshold $ASCENSIONTHRESHOLD = 1.0$.*

Proposition 13.1 (Fixed Point of Ω)**.** *The system admits a stable fixed point:*

$$\Omega^* = \frac{1-|\chi(I)|}{1-\lambda_{2,base}} = 2.0129 \tag{51}$$

for $\mathcal{C}^{\mathcal{R}}$ with $\chi(I) = 0$ and $\lambda_{2,base} = 0.5032$.

13.1 Stability Analysis

Proposition 13.2 (Asymptotic Stability)**.** *Ω^* is asymptotically stable with characteristic relaxation time:*

$$\tau_\Omega = \frac{1}{\min(k_1, k_2)} \approx 5.0 \tag{52}$$

Proof. The Jacobian at Ω^* is $\partial\dot{\Omega}/\partial\Omega|_{\Omega^*} = -\min(k_1, k_2) < 0$. The response to perturbations is exponential:

$$R(t) := \Omega(t) - \Omega^* = \delta\Omega \cdot e^{-t/\tau_\Omega} \tag{53}$$

□

13.2 Bifurcation

Proposition 13.3 (Saddle-Node Bifurcation)**.** *At $\|\mathcal{K}\| = 1$, the spectral gap vanishes ($\Delta\lambda = 0$), causing:*

$$\Omega \to \infty \tag{54}$$

This is a saddle-node bifurcation where the fixed point Ω^* disappears.

14 Validity Limits of $\mathcal{C}^{\mathcal{R}}$

14.1 Parameter Space Π

Definition 14.1 (Valid Parameter Space). *The validity domain of $\mathcal{C}^{\mathcal{R}}$ is:*

$$\Pi = \left\{ (K, \varepsilon, \kappa, \mathrm{inj}(I), \beth_q) \in \mathbb{R}^5 : \begin{array}{l} 0 \leq K < 1 \\ \varepsilon > 0 \\ \kappa > 0 \\ \mathrm{inj}(I) > K \cdot \max_u \|\Gamma(u)\| \\ 0.5 < \beth_q \leq 1 \end{array} \right\} \tag{55}$$

The dimension of Π is $n = 5$.

14.2 Critical Boundaries

Limit	Behavior	Regime
$K \to 1^-$	$\tau \to \infty$	Chaotic
$\mathrm{inj}(I) \to 0^+$	Stochastic projection	Collapse (Class 0)
$\Delta\lambda \to 0^+$	Multi-stability	Bifurcation
$\beth_q \to 0.5^+$	Weak closure	Transition

14.3 Non-Extension Theorem

Proposition 14.1 (Null Closure Inconsistency). *$\mathcal{C}^{\mathcal{R}}$ is inconsistent for systems with $\beth_q = 0$ (total informational transparency).*

Sketch. If $\beth_q = 0$, then mutual information $I(X;Y) = H(X)$, implying the internal state is a pure function of external input. The manifold $\mathcal{C}$ exerts no constraint, and the invariant attractor f^* vanishes. □

14.4 Phase Diagram

The parameter space decomposes into regions:

- $\mathcal{C}$: $K < 1, \kappa = 0$ (convex)
- $\mathcal{C}^{\mathcal{R}}$: $K < 1, \kappa > 0, \Delta\lambda > 0$ (critical)
- Chaotic: $K \geq 1$ or $\Delta\lambda \leq 0$
- Degenerate: $\beth_q \leq 0.5$

15 Second-Order Structure

15.1 Hessian of the Transition Operator

Definition 15.1 (Hessian)**.**

$$\text{Hess} = \nabla_\Psi(\nabla_\Psi \mathcal{T}_\varepsilon) = D^2\mathcal{P}_\varepsilon \circ (\mathcal{K} \otimes \mathcal{K}) \tag{56}$$

Proposition 15.1 (Hessian Signature)**.** *At the attractor,* Hess *has:*

- *Maximum eigenvalue:* $\lambda_{\text{Hess}} = -\|\mathcal{K}\|^2/r = -0.25$ *(for* $r = 1$*)*
- *Signature: semi-definite negative (stable curvature toward* $\mathcal{C}$*)*

15.2 Laplace-Beltrami Operator

On the manifold $\widetilde{\mathcal{C}} = S^1$ with radius r:

$$\sigma(\Delta_{LB}) = \left\{ \frac{n^2}{r^2} : n \in \mathbb{Z} \right\} = \{0, 1, 4, 9, 16, \ldots\} \tag{57}$$

Proposition 15.2 (Spectral Relation)**.** *The eigenvalues of the Jacobian relate to the Laplacian spectrum via:*

$$\lambda_n \approx \exp(-\Delta\lambda \cdot \sigma_n(\Delta_{LB})) \tag{58}$$

15.3 Fisher Information Metric

The Fisher metric on the parameter space is:

$$G_{\pi\pi} = \left(\tfrac{1}{1-\|\mathcal{K}\|}\right)^2 = 4.0 \quad \text{for } \|\mathcal{K}\| = 0.5 \tag{59}$$

15.4 Emergent Properties Unique to $\mathcal{C}^{\mathcal{R}}$

Property	$\mathcal{C}^{\mathcal{R}}$	$\mathcal{C}$
Geometric hysteresis $\int$ Hess $d\Psi$	Present	Absent
Euler zero $\chi(I) = 0$	Yes	No ($\chi = 1$)
Homotopy invariance $\pi_1 \sim= \mathbb{Z}$	Yes	No ($\pi_1 = 0$)
Phase resonance	Synchronized	Dissipated

A Counterexamples for Necessity

These counterexamples show that each hypothesis is necessary.

A.1 Without H1 (Convexity)

Let $\mathcal{C} = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (unit circle, not convex). For $\Psi = (0, 0)$, every point on $\mathcal{C}$ is equidistant. Projection is not unique.

A.2 Without H2 ($\|\mathcal{K}\| < 1$)

Let $\mathcal{K}$ be a rotation by angle $\theta \neq 0$ with $\|\mathcal{K}\| = 1$. Orbits cycle indefinitely; no fixed point exists.

A.3 Without H3 (Completeness)

Let $\mathcal{H} = C([0, 1], \mathbb{Q})$ (rational-valued continuous functions). Cauchy sequences may converge to irrational functions outside the space.

A.4 Without H4 (Compactness)

Let $\mathcal{C} = (0, 1)$ (open interval) and $\Gamma(u) = 1/u$. As $u \to 0$, attractors escape to infinity; $\mathcal{A}^*$ is not compact.

References

[1] anach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3, 133–181.

[2] reyszig, E. (1978). *Introductory Functional Analysis with Applications*. Wiley.

[3] udin, W. (1991). *Functional Analysis*. McGraw-Hill.

[4] rezis, H. (2011). *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Springer.